Kubernetes Ingress Nginx to F5 Nginx Ingress Migration

Why Ingress Nginx migration?

The community-maintained **Ingress NGINX controller for Kubernetes is retiring due to a lack of dedicated maintainers, significant security concerns, and the evolution of the Kubernetes networking model toward the more modern Gateway API**. It will reach end-of-life in March 2026, after which it will receive no further updates, bug fixes, or security patches.

The controller's flexibility, particularly the use of "snippets" annotations for injecting arbitrary NGINX configurations, created significant security risks and technical debt. Several critical vulnerabilities (e.g., CVE-2025-1974) were discovered that could not be easily addressed without major architectural changes, making the project a growing security liability.

The original Ingress API has limitations and cannot support modern networking requirements like advanced traffic splitting, multi-tenancy, and protocol-specific routing natively. The Kubernetes community is now aligned around the Gateway API, which offers a more expressive, extensible, and standardized approach to traffic management, with a clear separation of concerns for infrastructure and application teams.

Why F5 Nginx?

We are migrating because **Ingress-NGINX is annotation-driven, feature-limited, and nearing end-of-life**, while **F5 NGINX provides a structured, secure, enterprise-supported, and future-ready traffic management platform** that aligns with Kubernetes CRDs, Gateway API, and production security requirements.

The following table provides a comparison of key features:

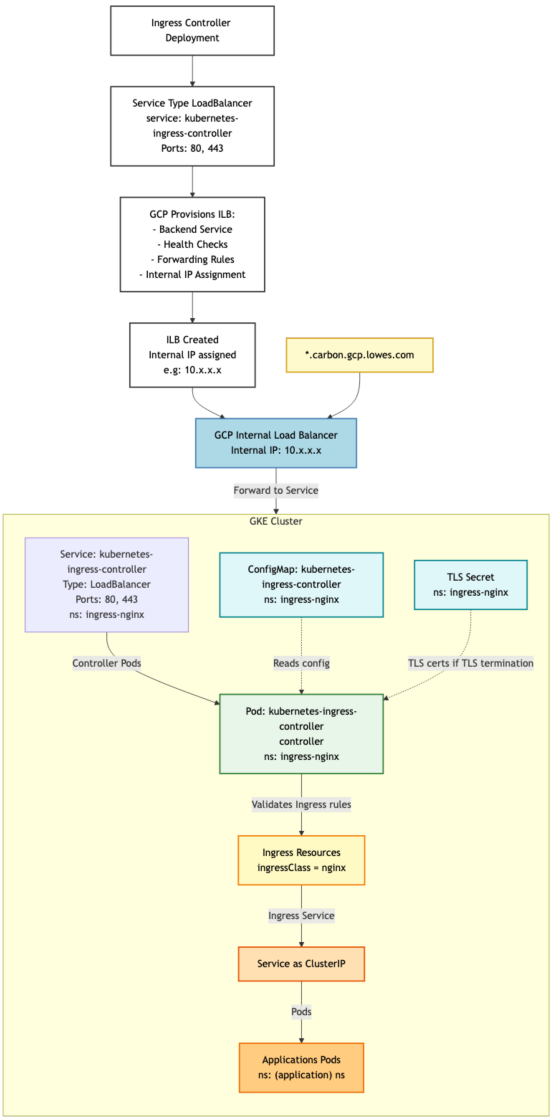
Feature/Aspect	Ingress Nginx	F5 NGINX Ingress Controller	Notes
Maintainer/Origin	Kubernetes Community (Open Source)	F5/NGINX Inc. (Open Source, Commercial)	
Primary Config for Advanced Features	Annotations	Custom Resources (CRDs)	CRDs offer structured, type-safe configuration.
Development Status	Ingress API “frozen”	Actively Developed, continuous feature releases	Allows continuous evolution.
Enterprise Support	Community support	24×7 Support available	Crucial for mission-critical applications.
Advanced Traffic Management	Limited, via annotations	Native CRD support (blue/green, canary, A/B testing, advanced load balancing, circuit breaking)	More robust, integrated capabilities.
Security Features	Basic TLS, some annotation-driven features	WAF (App Protect), mTLS, JWT/OIDC, DDoS, RBAC, SSL/TLS orchestration	Comprehensive security suite.
Protocol Support (beyond HTTP/S)	TCP/UDP via ConfigMap	Native CRD support (TransportServer for TCP/UDP, gRPC, WebSockets)	Unified traffic management for diverse applications.
Observability	Basic logging and metrics	Metrics, detailed logging, Prometheus/Grafana /OpenTelemetry integration	Enhanced troubleshooting and performance insights.
Configuration Complexity for Advanced Features	Annotation sprawl for complex features	Steeper initial CRD learning curve, but cleaner, more manageable, type-safe for complex configs	Favorable trade-off between immediate simplicity and long-term maintainability.
Cost	Free/Open Source	Free/Open Source and Commercial License (NGINX Plus)	Fit to needs, features, support, value

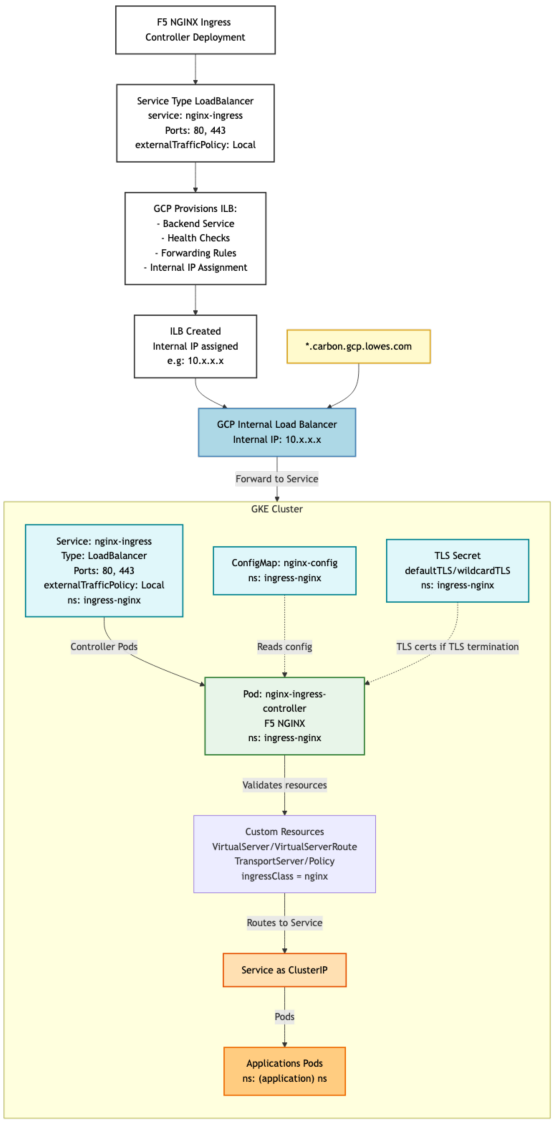
Architecture

Community Ingress-Nginx

F5 Nginx

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated





Installation and CRD-Based Configuration:

The NGINX Ingress Controller offers flexible installation options, including its official Helm chart. Ensure version consistency across the controller image, installation manifests, and documentation to prevent compatibility issues.

The new controller supports standard Kubernetes Ingress resources, but its power lies in its CRDs. This represents a significant difference in configurations compared to ingress-nginx.

Community Ingress-Nginx	**F5 NGINX Ingress Controller**	**Comparison Details**
Configuration Model	**Configuration Model**	
Relies heavily on annotations (nginx.ingress.kubernetes.io/rewrite-target).	Leverages Custom Resource Definitions (CRDs) like VirtualServer and Policy.	**CRD vs. Annotations:** CRDs provide a structured, type-safe, and Kubernetes-native approach to configuration.
Advanced Routing	**Advanced Routing**	
Uses annotations for features like canary deployments (nginx.ingress.kubernetes.io/canary-by-header).	Uses VirtualServer and VirtualServerRoute CRDs with native splits fields for weighted or header-based routing.	The CRD approach integrates advanced routing directly into the API, making it more declarative and GitOps-friendly.
Security Features	**Security Features**	

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

Limited by annotations; requires manual snippets for things like rate limiting.	Uses a dedicated Policy CRD for security features like WAF (with NGINX App Protect), mTLS, JWT/OIDC authentication, and fine-grained rate limiting.	The commercial controller offers a comprehensive and integrated security suite through the Policy CRD
TCP/UDP Support	**TCP/UDP Support**	
Managed via a ConfigMap, which can be less intuitive.	Uses a dedicated TransportServer CRD for TCP/UDP load balancing and TLS passthrough.	TransportServer offers a more structured and declarative way to manage non-HTTP traffic, aligning with the Kubernetes API model.

Converting Annotations: A critical step in this phase is converting your existing ingress-nginx annotations to their equivalents. Many common annotations have direct mappings, but for others, you might need to use NGINX snippets to replicate the desired behavior. The NGINX documentation can help guide you through this conversion process.

Understanding F5 Nginx Resources

Here is F5 Nginx resources which helps us to understand which to use when.

- Web app with https:// VirtualServer
- App where backend does SSL TransportServer + GlobalConfiguration
- Database/TCP/UDP service TransportServer + GlobalConfiguration
- Setting for ALL apps ConfigMap
- Rule for MULTIPLE apps Policy

VirtualServer - For Web Apps

- Use for HTTP/HTTPS (websites, APIs)
- 90% of use cases
- Examples: Jenkins, ForgeRock, APIs

TransportServer - For SSL Passthrough & TCP/UDP

- Backend handles SSL (not ingress)
- Non-HTTP protocols
- Examples: ArgoCD, databases, DNS

GlobalConfiguration - For TransportServer

- Required for TransportServer to work
- One per cluster
- Defines TCP/UDP listeners

ConfigMap - For Global Settings

- Applies to ALL apps
- Timeouts, buffers, security
- Set once, affects everything

Policy - For Reusable Rules

- Authentication, rate limiting
- Define once, use many times
- Examples: Basic auth, IP allowlist

Annotations Summary in F5 Nginx Support:

Summary of the available annotations to use standrad ingress in F5 Nginx

Features	Ingress Nginx	F5 Standard Ingress	Status for F5 OSS
Session Affinity	nginx.ingress.kubernetes.io/affinity	Kubernetes Service sessionAffinity	Alternative as Service
Basic Auth Secret	nginx.ingress.kubernetes.io/auth-secret	nginx.org/basic-auth-secret	Available
Auth Signin	nginx.ingress.kubernetes.io/auth-signin	Custom snippets	Custom
Auth Type	nginx.ingress.kubernetes.io/auth-type	nginx.org/basic-auth-secret	Available
Auth URL	nginx.ingress.kubernetes.io/auth-url	Custom snippets	Custom
Backend Protocol	nginx.ingress.kubernetes.io/backend-protocol	nginx.org/ssl-services, nginx.org/grpc-services	Available

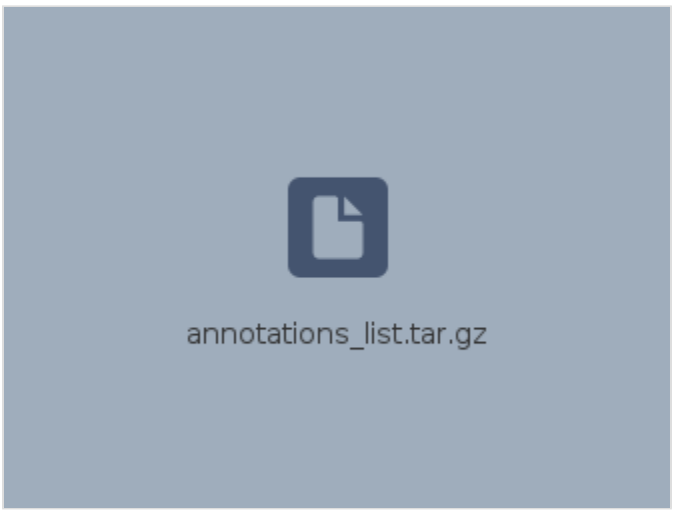
All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

Body Size	nginx.ingress.kubernetes.io/proxy-body-size	nginx.org/client-max-body-size	Available
Configuration Snippet	nginx.ingress.kubernetes.io/configuration-snippet	nginx.org/location-snippets	Available
Enable CORS	nginx.ingress.kubernetes.io/enable-cors	Custom snippets	Custom
Underscores in Headers	nginx.ingress.kubernetes.io/enable-underscores-in-headers	ConfigMap global	Global
Force SSL Redirect	nginx.ingress.kubernetes.io/force-ssl-redirect	nginx.org/redirect-to-https	Available
Large Client Headers	nginx.ingress.kubernetes.io/large-client-header-buffers	ConfigMap global	Global
Proxy Body Size	nginx.ingress.kubernetes.io/proxy-body-size	nginx.org/client-max-body-size	Available
Proxy Buffer Size	nginx.ingress.kubernetes.io/proxy-buffer-size	nginx.org/proxy-buffer-size	Available
Proxy Buffers Number	nginx.ingress.kubernetes.io/proxy-buffers-number	nginx.org/proxy-buffers	Available
Proxy Read Timeout	nginx.ingress.kubernetes.io/proxy-read-timeout	nginx.org/proxy-read-timeout	Available
Proxy Send Timeout	nginx.ingress.kubernetes.io/proxy-send-timeout	nginx.org/proxy-send-timeout	Available
Rewrite Target	nginx.ingress.kubernetes.io/rewrite-target	nginx.org/rewrites	Available
Send Timeout	nginx.ingress.kubernetes.io/send-timeout	nginx.org/proxy-send-timeout	Available
Session Cookie Hash	nginx.ingress.kubernetes.io/session-cookie-hash	Kubernetes Service sessionAffinity	Alternative as Service
Session Cookie Name	nginx.ingress.kubernetes.io/session-cookie-name	Kubernetes Service sessionAffinity	Alternative as Service
SSL Passthrough	nginx.ingress.kubernetes.io/ssl-passthrough	TransportServer CRD	Different
SSL Redirect	nginx.ingress.kubernetes.io/ssl-redirect	nginx.org/redirect-to-https	Available
Use Forwarded Headers	nginx.ingress.kubernetes.io/use-forwarded-headers	ConfigMap global	Global
Use Regex	nginx.ingress.kubernetes.io/use-regex	nginx.org/path-regex	Available
Websocket Services	nginx.ingress.kubernetes.io/websocket-services	nginx.org/websocket-services	Available

List of Annotations used across Carbon consumers

- ingress.kubernetes.io/affinity
- ingress.kubernetes.io/auth-secret
- ingress.kubernetes.io/auth-signin
- ingress.kubernetes.io/auth-type
- ingress.kubernetes.io/auth-url
- ingress.kubernetes.io/backend-protocol
- ingress.kubernetes.io/body-size
- ingress.kubernetes.io/configuration-snippet
- ingress.kubernetes.io/enable-cors
- ingress.kubernetes.io/enable-underscores-in-headers
- ingress.kubernetes.io/force-ssl-redirect
- ingress.kubernetes.io/large-client-header-buffers
- ingress.kubernetes.io/proxy-body-size
- ingress.kubernetes.io/proxy-buffer-size
- ingress.kubernetes.io/proxy-buffers-number
- ingress.kubernetes.io/proxy-read-timeout
- ingress.kubernetes.io/proxy-send-timeout
- ingress.kubernetes.io/rewrite-target
- ingress.kubernetes.io/send-timeout
- ingress.kubernetes.io/session-cookie-hash
- ingress.kubernetes.io/session-cookie-name
- ingress.kubernetes.io/ssl-passthrough
- ingress.kubernetes.io/ssl-redirect
- ingress.kubernetes.io/use-forwarded-headers
- ingress.kubernetes.io/use-regex
- ingress.kubernetes.io/websocket-services

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated



How to use the manifest in F5 Nginx

Service with session affinity

Service with session Affinity

```
apiVersion: v1
kind: Service
metadata:
  name: backend-service
  namespace: default
spec:
  sessionAffinity: ClientIP
  sessionAffinityConfig:
    clientIP:
      timeoutSeconds: 10800  # 3 hours (default is 10800)
  type: ClusterIP
  ports:
  - name: http
    port: 80
    targetPort: 8080
    protocol: TCP
  selector:
    app: backend
    version: v1
```

Configmap Global

- Global NGINX configuration
- Proxy settings, SSL/TLS, logging, headers

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

ConfigMap Global

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: nginx-config
  namespace: nginx-ingress
data:
  proxy-connect-timeout: "60s"
  proxy-read-timeout: "60s"
  proxy-send-timeout: "60s"
  client-max-body-size: "10m"
  client-body-buffer-size: "16k"
  client-header-buffer-size: "1k"
  large-client-header-buffers: "4 8k"
  proxy-buffer-size: "8k"
  proxy-buffers: "8 8k"
  proxy-busy-buffers-size: "16k"
  keepalive-timeout: "75s"
  keepalive-requests: "100"
  upstream-keepalive-connections: "32"
  upstream-keepalive-timeout: "60s"
  ssl-protocols: "TLSv1.2 TLSv1.3"
  ssl-ciphers: "HIGH:!aNULL:!MD5"
  ssl-prefer-server-ciphers: "true"
  http2: "true"
  log-format: '$remote_addr - $remote_user [$time_local] "$request" $status $body_bytes_sent "$http_referer"
"$http_user_agent" "$http_x_forwarded_for"'
  access-log-off: "false"
  error-log-level: "notice"
  real-ip-header: "X-Forwarded-For"
  set-real-ip-from: "0.0.0.0/0"
  real-ip-recursive: "true"
  proxy-hide-headers: "X-Powered-By,Server"
  server-tokens: "false"
  underscores-in-headers: "true"
  use-forwarded-headers: "true"
  worker-processes: "auto"
  worker-connections: "10240"
  server-names-hash-bucket-size: "256"
  server-names-hash-max-size: "1024"
  variables-hash-bucket-size: "256"
  variables-hash-max-size: "1024"
```

Virtual Server

- VirtualServer with health checks
- Load balancing, timeouts, buffers
- Multiple routes and upstreams
- Rate limiting policies
- Path rewrites, custom headers

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

Virtual Server basic

```
apiVersion: k8s.nginx.org/v1
kind: VirtualServer
metadata:
  name: basic-virtualserver
  namespace: default
spec:
  host: app.example.com
  tls:
    secret: app-tls-secret
    redirect:
      enable: true  # Redirect HTTP to HTTPS
  upstreams:
  - name: backend
    service: backend-service
    port: 80
    lb-method: round_robin
    keepalive: 32
    healthCheck:
      enable: true
      path: /health
      interval: 20s
      timeout: 5s
      fails: 3
      passes: 2
    connect-timeout: 30s
    read-timeout: 60s
    send-timeout: 60s
    client-max-body-size: 10m
  routes:
  - path: /
    action:
      pass: backend
```

VirtualServer Advanced with Multiple Routes and Features

Virtual Server Advanced

```
# Advanced VirtualServer with Multiple Routes and Features

apiVersion: k8s.nginx.org/v1
kind: VirtualServer
metadata:
  name: advanced-virtualserver
  namespace: default
spec:
  host: api.example.com
  tls:
    secret: api-tls-secret
    redirect:
      enable: true
      code: 301
  upstreams:
  - name: api-v1
    service: api-v1-service
    port: 8080
    lb-method: least_conn
    keepalive: 64
    max-fails: 3
    fail-timeout: 10s
    connect-timeout: 30s
    read-timeout: 300s
    send-timeout: 300s
    client-max-body-size: 50m
  - name: api-v2
```

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

```
    service: api-v2-service
    port: 8080
    lb-method: least_conn
    keepalive: 64
  - name: websocket-backend
    service: websocket-service
    port: 8080
    lb-method: ip_hash
  routes:
  - path: /api/v1
    action:
      pass: api-v1
  - path: /api/v2
    policies:
    - name: rate-limit-policy
    action:
      pass: api-v2
  - path: /ws
    action:
      pass: websocket-backend
  - path: /old-path
    action:
      pass: api-v1
      rewrite: /new-path
  - path: /secure
    action:
      pass: api-v1
    responseHeaders:
      add:
      - name: X-Frame-Options
        value: DENY
      - name: X-Content-Type-Options
        value: nosniff
      hide:
      - X-Powered-By
---
apiVersion: k8s.nginx.org/v1
kind: Policy
metadata:
  name: rate-limit-policy
  namespace: default
spec:
  rateLimit:
    rate: 10r/s
    key: ${binary_remote_addr}
    zoneSize: 10M
```

For gRPC

- gRPC-specific configuration
- Long-running stream support
- gRPC health checks

VirtualServer for gRPC Services

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

Virtual Server for gRPC

```
apiVersion: k8s.nginx.org/v1
kind: VirtualServer
metadata:
  name: grpc-virtualserver
  namespace: default
spec:
  host: grpc.example.com
  tls:
    secret: grpc-tls-secret
  upstreams:
  - name: grpc-backend
    service: grpc-service
    port: 50051
    type: grpc
  routes:
  - path: /
    action:
      pass: grpc-backend
```

Virtual Server with snippets

- Custom NGINX snippets for CORS
- External authentication with auth_request
- Custom logging

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

Virtual Server with Snippets

```
apiVersion: k8s.nginx.org/v1
kind: VirtualServer
metadata:
  name: virtualserver-with-snippets
  namespace: default
spec:
  host: custom.example.com
  tls:
    secret: custom-tls-secret
  upstreams:
  - name: backend
    service: backend-service
    port: 80
  routes:
  - path: /api
    location-snippets: |
      if ($request_method = 'OPTIONS') {
        add_header 'Access-Control-Allow-Origin' '$http_origin' always;
        add_header 'Access-Control-Allow-Methods' 'GET, POST, PUT, DELETE, OPTIONS' always;
        add_header 'Access-Control-Allow-Headers' 'Authorization, Content-Type, X-Requested-With' always;
        add_header 'Access-Control-Max-Age' 1728000 always;
        add_header 'Content-Length' 0;
        return 204;
      }
      add_header 'Access-Control-Allow-Origin' '$http_origin' always;
      add_header 'Access-Control-Allow-Credentials' 'true' always;
    action:
      pass: backend
  - path: /auth
    location-snippets: |
      auth_request /validate;
      auth_request_set $auth_status $upstream_status;
      location = /validate {
        internal;
        proxy_pass http://auth-service.default.svc.cluster.local/validate;
        proxy_pass_request_body off;
        proxy_set_header Content-Length "";
        proxy_set_header X-Original-URI $request_uri;
        proxy_set_header X-Original-Method $request_method;
      }
    action:
      pass: backend
    server-snippets: |
    # Custom logging format
    log_format custom_format '$remote_addr - $remote_user [$time_local] '
                             '"$request" $status $body_bytes_sent '
                             '"$http_referer" "$http_user_agent" '
                             'rt=$request_time';
    access_log /var/log/nginx/custom-access.log custom_format;
```

Transport Server Resources:

Transportserver-ssl-passthrough

- SSL/TLS passthrough (no termination)
- SNI-based routing
- GlobalConfiguration for listener

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

Transport Server SSL

```
apiVersion: k8s.nginx.org/v1
kind: TransportServer
metadata:
  name: ssl-passthrough-app
  namespace: default
spec:
  listener:
    name: tls-passthrough
    protocol: TLS_PASSTHROUGH
  host: secure.example.com
  upstreams:
  - name: secure-backend
    service: secure-app-service
    port: 8443
    healthCheck:
      enable: true
      timeout: 5s
      interval: 20s
      port: 8443
  action:
    pass: secure-backend
```

Transport server with TCP

- TCP load balancing (e.g., PostgreSQL)
- TCP health checks
- Non-HTTP protocol support

Transport Server in TCP

```
apiVersion: k8s.nginx.org/v1
kind: TransportServer
metadata:
  name: tcp-database
  namespace: default
spec:
  listener:
    name: tcp-database
    protocol: TCP
  upstreams:
  - name: postgres-backend
    service: postgres-service
    port: 5432
  action:
    pass: postgres-backend
---
apiVersion: k8s.nginx.org/v1
kind: GlobalConfiguration
metadata:
  name: nginx-configuration
  namespace: nginx-ingress
spec:
  listeners:
  - name: tcp-database
    port: 5432
    protocol: TCP
```

Global Configuration:

GlobalConfiguration resource to define the global configuration parameters of F5 NGINX Ingress Controller. The resource supports configuring listeners for TCP and UDP load balancing, and is implemented as a Custom resource. Listeners are required by TransportServer resources and can be used to configure custom listeners for VirtualServers.

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

Note: The GlobalConfiguration resource acts as the source of truth for all listeners in that cluster. If a consumer needs to add a new port, they must update that existing resource rather than creating a second one, as NGINX will only ever look at the one specified in your startup arguments.

Please always create metadata name as f5-nginx-global-configuration in f5-nginx-ingress namespace always.

GlobalConfiguration

```
apiVersion: k8s.nginx.org/v1
kind: GlobalConfiguration
metadata:
  name: f5-nginx-global-configuration
  namespace: f5-nginx-ingress
spec:
  listeners:
    - name: http-8088
      port: 8088
      protocol: HTTP
    - name: tcp-9000
      port: 9000
      protocol: TCP
```

Policy Resources

- Rate limiting policy
- IP whitelist/access control
- Basic authentication
- Example VirtualServer using policies

Policy Resources

```
apiVersion: k8s.nginx.org/v1
kind: Policy
metadata:
  name: rate-limit-api
  namespace: default
spec:
  rateLimit:
    rate: 100r/s
    key: ${binary_remote_addr}
    zoneSize: 10M
    delay: 50
    noDelay: false
    burst: 200
    dryRun: false
---
apiVersion: k8s.nginx.org/v1
kind: Policy
metadata:
  name: ip-whitelist
  namespace: default
spec:
  accessControl:
    allow:
    - 10.0.0.0/8
    - 192.168.1.0/24
    deny:
    - all
---
apiVersion: k8s.nginx.org/v1
kind: Policy
metadata:
  name: basic-auth
  namespace: default
spec:
  basicAuth:
    secret: basic-auth-secret
```

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

```
    realm: "Protected Area"
---
apiVersion: v1
kind: Secret
metadata:
  name: basic-auth-secret
  namespace: default
type: Opaque
data:
  # Format: username:password_hash (base64 encoded)
  # Example: admin:$apr1$xyz... (use htpasswd to generate)
  auth: YWRtaW46JGFwcjEkLi4uCg==
---
apiVersion: k8s.nginx.org/v1
kind: VirtualServer
metadata:
  name: virtualserver-with-policies
  namespace: default
spec:
  host: protected.example.com
  tls:
    secret: protected-tls-secret
  policies:
  - name: ip-whitelist
  upstreams:
  - name: backend
    service: backend-service
    port: 80
  routes:
  - path: /public
    action:
      pass: backend
  - path: /api
    policies:
    - name: rate-limit-api
    - name: basic-auth
    action:
      pass: backend
```

Default Values:

Values

```
This document lists the default settings for F5 NGINX Ingress Controller (Open Source) to help you
understand what's configured out-of-the-box and what you need to customize.

## Connection & Timeout Defaults

| Setting | Default Value | Description | When to Change |
|---------|--------------|-------------|----------------|
| `proxy-connect-timeout` | **60s** | Time to establish connection to backend | Slow backends, increase to
120s |
| `proxy-read-timeout` | **60s** | Time to read response from backend | Long-running requests (uploads,
reports) |
| `proxy-send-timeout` | **60s** | Time to send request to backend | Large request bodies |
| `keepalive-timeout` | **75s** | Client connection keepalive | Adjust based on client behavior |
| `upstream-keepalive-connections` | **32** | Persistent connections to backends | High traffic, increase to
64-128 |
| `upstream-keepalive-timeout` | **60s** | Backend connection keepalive | Match backend timeout settings |

## Buffer Defaults

| Setting | Default Value | Description | When to Change |
|---------|--------------|-------------|----------------|
| `proxy-buffer-size` | **8k** | Buffer for response headers | OAuth/SAML with large tokens |
| `proxy-buffers` | **8 4k** | Number and size of buffers | Large responses, increase to 8 8k |
| `proxy-busy-buffers-size` | **16k** | Size of busy buffers | High throughput scenarios |
```

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

`client-body-buffer-size`	**16k**	Buffer for request body	Large POST requests
`client-header-buffer-size`	**1k**	Buffer for request headers	Standard headers fit in 1k
`large-client-header-buffers`	**4 8k**	Large header buffers	OAuth tokens, cookies

Client Request Defaults

Setting	Default Value	Description	When to Change
`client-max-body-size`	**1m**	Maximum request body size	File uploads, increase to 10m-100m
`client-body-timeout`	**60s**	Time to read client request body	Slow clients, large uploads
`client-header-timeout`	**60s**	Time to read client headers	Rarely needs changing

Upstream/Backend Defaults

Setting	Default Value	Description	When to Change
`lb-method`	**round_robin**	Load balancing algorithm	Use least_conn for stateful apps
`max-fails`	**1**	Failed attempts before marking unhealthy	Increase for flaky backends
`fail-timeout`	**10s**	Time to mark backend as unavailable	Adjust based on recovery time
`max-conns`	**0** (unlimited)	Max connections per backend	Protect backends from overload

Available load balancing methods:
- `round_robin` - Default, equal distribution
- `least_conn` - Send to backend with fewest connections (recommended for stateful apps)
- `ip_hash` - Session affinity based on client IP
- `hash` - Custom hash-based routing
- `random` - Random selection

TLS/SSL Defaults

Setting	Default Value	Description	When to Change
`ssl-protocols`	**TLSv1.2 TLSv1.3**	Allowed TLS versions	Security policy requirements
`ssl-prefer-server-ciphers`	**false**	Server cipher preference	Enable for strict security
`ssl-ciphers`	**HIGH:!aNULL:!MD5**	Allowed cipher suites	Compliance requirements
`ssl-redirect`	**false**	Force HTTPS redirect	Enable for production
`hsts`	**false**	HTTP Strict Transport Security	Enable for production
`hsts-max-age`	**2592000** (30 days)	HSTS max age	Increase to 31536000 (1 year)

HTTP/2 & WebSocket Defaults

Setting	Default Value	Description	When to Change
`http2`	**true**	HTTP/2 support	Rarely disable
WebSocket upgrade	**Automatic**	Upgrades HTTP to WebSocket	No config needed

Health Check Defaults

Setting	Default Value	Description	When to Change
`healthCheck.enable`	**false**	Active health checks	Enable for critical services
`healthCheck.interval`	**5s**	Check interval	Adjust based on backend load
`healthCheck.fails`	**1**	Failures before unhealthy	Increase for flaky backends
`healthCheck.passes`	**1**	Successes before healthy	Increase for stability
`healthCheck.uri`	**/**	Health check path	Set to actual health endpoint
`healthCheck.port`	Same as upstream	Health check port	Use different port if needed

Important: Active health checks in F5 NGINX Open Source are **passive only** by default. For active health checks, you need to explicitly configure them.

VirtualServer
- Default: `round_robin`
- No session affinity

Service (Kubernetes)
- Default: `sessionAffinity: None`

Session Affinity Options

Kubernetes Service Level:
```yaml

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated
```

```
sessionAffinity: ClientIP
sessionAffinityConfig:
 clientIP:
 timeoutSeconds: 10800 # Default: 3 hours
```

F5 NGINX Plus Only (not available in Open Source):
- Cookie-based session affinity
- JWT-based routing
- Active health checks

Logging Defaults

Setting	Default Value	Description	When to Change
`access-log-off`	**false**	Disable access logs	Disable for high traffic
`error-log-level`	**notice**	Error log verbosity	Use `debug` for troubleshooting
`log-format`	**combined**	Access log format	Customize for observability

Log levels: `debug`, `info`, `notice`, `warn`, `error`, `crit`

Security Defaults

Setting	Default Value	Description	When to Change
`server-tokens`	**true**	Show NGINX version	Disable for security
`underscores-in-headers`	**false**	Allow underscores in headers	Enable for telemetry (file 21, 24)
`use-forwarded-headers`	**false**	Trust X-Forwarded-* headers	Enable behind load balancer (file 21, 24)
`real-ip-header`	**X-Real-IP**	Header for real client IP	Use X-Forwarded-For behind LB
`set-real-ip-from`	**None**	Trusted proxy CIDRs	Set to LB IP ranges

Rate Limiting Defaults

Setting	Default Value	Description	When to Change
Rate limiting	**Disabled**	No rate limits by default	Enable via Policy resource
`limit-req-status-code`	**429**	Status for rate limit	Standard HTTP 429
`limit-conn-status-code`	**429**	Status for connection limit	Standard HTTP 429

Worker Process Defaults

Setting	Default Value	Description	When to Change
`worker-processes`	**auto**	Number of worker processes	Matches CPU cores
`worker-connections`	**1024**	Max connections per worker	High traffic, increase to 10240
`worker-rlimit-nofile`	**65536**	Max open files	Very high traffic

Hash Bucket Defaults

Setting	Default Value	Description	When to Change
`server-names-hash-bucket-size`	**64**	Server name hash bucket	Many hosts, increase to 128-256
`server-names-hash-max-size`	**512**	Server name hash table	Many hosts, increase to 1024
`variables-hash-bucket-size`	**64**	Variable hash bucket	Many variables
`variables-hash-max-size`	**512**	Variable hash table	Many variables

gRPC-Specific Defaults

Setting	Default Value	Description	When to Change
`upstream.type`	**http**	Backend protocol	Set to `grpc` for gRPC (file 21)
gRPC timeouts	**60s**	Standard timeouts	Increase to 3600s for streaming

Configuration Precedence

Settings can be configured at multiple levels, with this precedence (highest to lowest):

1. **VirtualServer `location-snippets`** - Per-route custom NGINX directives
2. **VirtualServer `upstream` settings** - Per-upstream configuration
3. **VirtualServer `spec` settings** - Per-VirtualServer configuration

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

```
4. **ConfigMap** - Global settings for all resources
5. **Built-in defaults** - F5 NGINX compiled defaults

## Quick Reference: Common Overrides

```yaml
# Global ConfigMap (applies to all VirtualServers)
apiVersion: v1
kind: ConfigMap
metadata:
 name: nginx-config
 namespace: nginx-ingress
data:
 proxy-connect-timeout: "60s"
 proxy-read-timeout: "60s"
 client-max-body-size: "10m"
 proxy-buffer-size: "8k"
 server-tokens: "false"
 underscores-in-headers: "false"
 use-forwarded-headers: "false"

# Per-VirtualServer (overrides global)
apiVersion: k8s.nginx.org/v1
kind: VirtualServer
spec:
 upstreams:
 - name: backend
 connect-timeout: 120s # Override global
 read-timeout: 300s # Override global
 client-max-body-size: 50m # Override global
```
```

Community Ingress-Nginx Vs F5 Nginx

Community Nginx Basic

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  annotations:
    kubernetes.io/ingress.class: nginx
    nginx.ingress.kubernetes.io/backend-protocol:
http
  name: jenkins-ingress
  namespace: jenkins
spec:
  rules:
  - host: jenkins.carbon.gcp.lowes.com
    http:
      paths:
      - backend:
          service:
            name: carbon-jenkins
            port:
              number: 8080
        path: /
        pathType: Prefix
  tls:
  - hosts:
    - jenkins.carbon.gcp.lowes
```

F5 Nginx Basic

```
apiVersion: k8s.nginx.org/v1
kind: VirtualServer
metadata:
  name: jenkins-ingress
  namespace: jenkins
spec:
  host: jenkins.carbon.gcp.lowes.com
  tls:
    secret: jenkins-tls
  upstreams:
  - name: jenkins-backend
    service: carbon-jenkins
    port: 8080
   routes:
  - path: /
    action:
      pass: jenkins-backend
```

F5 Nginx Annotations with Cookies, Session Affinity

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

Community Nginx Annotations like Cookies, Session Affinity

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  annotations:
    nginx.ingress.kubernetes.io/affinity: cookie
    nginx.ingress.kubernetes.io/body-size: 64m
    nginx.ingress.kubernetes.io/configuration-
snippet: |
      add_header X-ForgeRock-TransactionId $req_id;
      proxy_set_header X-ForgeRock-TransactionId
$req_id;
    nginx.ingress.kubernetes.io/enable-cors:
"false"
    nginx.ingress.kubernetes.io/proxy-body-size:
64m
    nginx.ingress.kubernetes.io/proxy-buffer-size:
16k
    nginx.ingress.kubernetes.io/proxy-read-
timeout: "600"
    nginx.ingress.kubernetes.io/proxy-send-
timeout: "600"
    nginx.ingress.kubernetes.io/rewrite-target: /am
/$2
    nginx.ingress.kubernetes.io/send-timeout: "600"
    nginx.ingress.kubernetes.io/session-cookie-
hash: sha1
    nginx.ingress.kubernetes.io/session-cookie-
name: route
    nginx.ingress.kubernetes.io/ssl-redirect:
"true"
    nginx.ingress.kubernetes.io/websocket-
services: rcs-agent
    nginx.org/websocket-services: rcs-agent
  name: forgerock
  namespace: commerce-identity-ns
spec:
  ingressClassName: nginx
  rules:
  - host: prod.am.identity.gcp.lowes.com
    http:
      paths:
      - backend:
          service:
            name: am
            port:
              number: 8080
        path: /openam(/|$)(.*)
        pathType: Prefix
  tls:
  - hosts:
    - prod.am.identity.gcp.lowes.com
    secretName: nginx-cert-secret
```

```
apiVersion: v1
kind: Service
metadata:
  name: am
  namespace: commerce-identity-ns
spec:
  type: ClusterIP
  sessionAffinity: ClientIP
  sessionAffinityConfig:
    clientIP:
      timeoutSeconds: 10800
  ports:
  - name: http
    port: 8080
    targetPort: 8080
  selector:
    app: am

---
# F5 NGINX VirtualServer Configuration
apiVersion: k8s.nginx.org/v1
kind: VirtualServer
metadata:
  name: forgerock
  namespace: commerce-identity-ns
spec:
  host: prod.am.identity.gcp.lowes.com
  tls:
    secret: nginx-cert-secret
    redirect:
      enable: true  # Redirects HTTP to HTTPS
      code: 301
  upstreams:
  - name: am-backend
    service: am
    port: 8080
    lb-method: least_conn
    keepalive: 64
    connect-timeout: 30s
    read-timeout: 600s
    send-timeout: 600s
    client-max-body-size: 64m
    buffering: true
    buffers:
      number: 8
      size: 16k
  routes:
  - path: /openam
    action:
      pass: am-backend
    location-snippets: |
      rewrite ^/openam(/|$)(.*) /am/$2 break;
      add_header X-ForgeRock-TransactionId
$request_id always;
      proxy_set_header X-ForgeRock-TransactionId
$request_id;
      proxy_http_version 1.1;
      proxy_set_header Upgrade $http_upgrade;
      proxy_set_header Connection "upgrade";

---
# Notes:
# This configuration replaces the following
original annotations:
# - nginx.ingress.kubernetes.io/affinity: cookie
# - nginx.ingress.kubernetes.io/session-cookie-
name: route
# - nginx.ingress.kubernetes.io/session-cookie-
hash: sha1
# - nginx.ingress.kubernetes.io/body-size: 64m
# - nginx.ingress.kubernetes.io/proxy-body-size:
```

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

```
64m
# - nginx.ingress.kubernetes.io/proxy-buffer-size:
16k
# - nginx.ingress.kubernetes.io/proxy-read-
timeout: "600"
# - nginx.ingress.kubernetes.io/proxy-send-
timeout: "600"
# - nginx.ingress.kubernetes.io/send-timeout: "600"
# - nginx.ingress.kubernetes.io/ssl-redirect:
"true"
# - nginx.ingress.kubernetes.io/rewrite-target: /am
/$2
# - nginx.ingress.kubernetes.io/configuration-
snippet: (custom headers)
# - nginx.ingress.kubernetes.io/enable-cors:
"false"
# - nginx.ingress.kubernetes.io/websocket-
services: rcs-agent

---
# Backend Service with Session Affinity
# Replaces: nginx.ingress.kubernetes.io/affinity:
cookie
# Replaces: nginx.ingress.kubernetes.io/session-
cookie-name: route
# Replaces: nginx.ingress.kubernetes.io/session-
cookie-hash: sha1
# Note: F5 NGINX open source doesn't support
cookie-based affinity, Using Kubernetes native
ClientIP session affinity instead
# Session Cookie feature is supported only in
NGINX Plus.
```

F5 Nginx Annotation Rewrite

```
apiVersion: v1
kind: Service
metadata:
  name: am
  namespace: commerce-identity-ns
spec:
  type: ClusterIP
  sessionAffinity: ClientIP
  sessionAffinityConfig:
    clientIP:
      timeoutSeconds: 10800
  ports:
  - name: http
    port: 8080
    targetPort: 8080
  selector:
    app: am
# Note: F5 NGINX open source doesn't support
cookie-based affinity, Using Kubernetes native
ClientIP session affinity instead
# Session Cookie feature is supported only in
NGINX Plus.
---
apiVersion: k8s.nginx.org/v1
kind: VirtualServer
metadata:
  name: am-admin-rewrite
  namespace: commerce-identity-ns
spec:
  host: dev.am.identity.gcp.lowes.com
  tls:
```

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

Community Nginx Rewrite

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  annotations:
    nginx.ingress.kubernetes.io/proxy-body-size:
64m
    nginx.ingress.kubernetes.io/proxy-buffer-size:
16k
    nginx.ingress.kubernetes.io/proxy-read-
timeout: "600"
    nginx.ingress.kubernetes.io/proxy-send-
timeout: "600"
    nginx.ingress.kubernetes.io/rewrite-target: /am
/oauth2/realms/root/realms/lowes/.well-known
/openid-configuration
    nginx.ingress.kubernetes.io/send-timeout: "600"
    nginx.ingress.kubernetes.io/session-cookie-
hash: sha1
    nginx.ingress.kubernetes.io/session-cookie-
name: route
    nginx.ingress.kubernetes.io/ssl-redirect:
"true"
    nginx.ingress.kubernetes.io/use-regex: "true"
    nginx.ingress.kubernetes.io/websocket-
services: rcs-agent
    nginx.org/websocket-services: rcs-agent
  name: am-admin-rewrite
  namespace: commerce-identity-ns
spec:
  ingressClassName: nginx
  rules:
  - host: dev.am.identity.gcp.lowes.com
    http:
      paths:
      - backend:
          service:
            name: am
            port:
              number: 8080
        path: /am/oauth2/realms/root/realms/lowes/.
well-known/oauth-authorization-server
        pathType: ImplementationSpecific
  tls:
  - hosts:
    - dev.am.identity.gcp.lowes.com
    secretName: nginx-cert-secret
    secret: nginx-cert-secret
    redirect:
      enable: true
      code: 301
  upstreams:
  - name: am-backend
    service: am
    port: 8080
    connect-timeout: 60s
    read-timeout: 600s
    send-timeout: 600s
    client-max-body-size: 64m
    buffering: true
    buffers:
      number: 4
      size: 8k
    buffer-size: 16k
  routes:
  - path: /am/oauth2/realms/root/realms/lowes/.
well-known/oauth-authorization-server
    action:
      pass: am-backend
    location-snippets: |
      rewrite ^/am/oauth2/realms/root/realms
/lowes/.well-known/oauth-authorization-server$ /am
/oauth2/realms/root/realms/lowes/.well-known
/openid-configuration break;
      proxy_http_version 1.1;
      proxy_set_header Upgrade $http_upgrade;
      proxy_set_header Connection "upgrade";
---
#Note:
# TLS Configuration
# Replaces: nginx.ingress.kubernetes.io/ssl-
redirect: "true"
# Timeouts
# Replaces: nginx.ingress.kubernetes.io/proxy-read-
timeout: "600"
# Replaces: nginx.ingress.kubernetes.io/proxy-send-
timeout: "600"
# Replaces: nginx.ingress.kubernetes.io/send-
timeout: "600"
# Note: No proxy-connect-timeout annotation = uses
NGINX default 60s
# Body size limit
# Replaces: nginx.ingress.kubernetes.io/proxy-body-
size: 64m
# Buffer settings
# Replaces: nginx.ingress.kubernetes.io/proxy-
buffer-size: 16k
# Replaces: nginx.ingress.kubernetes.io/rewrite-
target
# Replaces: nginx.ingress.kubernetes.io/use-regex:
"true"
# Replaces: nginx.ingress.kubernetes.io/websocket-
services: rcs-agent
```

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

Community Nginx Basic Auth

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  annotations:
    ingress.kubernetes.io/ssl-passthrough: "false"
    kubernetes.io/ingress.class: nginx
    nginx.ingress.kubernetes.io/auth-secret:
schema-auth-secret
    nginx.ingress.kubernetes.io/auth-type: basic
    nginx.ingress.kubernetes.io/backend-protocol:
HTTPS
    nginx.ingress.kubernetes.io/ssl-passthrough:
"false"
  name: kite-schema-registry-admin-ingress
  namespace: kite-schema-registry-ns
spec:
  rules:
  - host: kite-schema-registry-admin.central1.kite.
carbon-stg01.gcp.lowes.com
    http:
      paths:
      - backend:
          service:
            name: kite-schema-registry-svc
            port:
              number: 8081
        path: /
        pathType: ImplementationSpecific
  tls:
  - hosts:
    - kite-schema-registry-admin.central1.kite.
carbon-stg01.gcp.lowes.com
    secretName: kite-schema-registry-ext-listener-
cert
```

F5 Nginx Basic Auth

```
apiVersion: k8s.nginx.org/v1
kind: VirtualServer
metadata:
  name: kite-schema-registry-admin-ingress
  namespace: kite-schema-registry-ns
spec:
  host: kite-schema-registry-admin.central1.kite.
carbon-stg01.gcp.lowes.com
  tls:
    secret: kite-schema-registry-ext-listener-cert
  policies:
  - name: basic-auth-policy
  upstreams:
  - name: schema-registry-backend
    service: kite-schema-registry-svc
    port: 8081
    tls:
      enable: true
  routes:
  - path: /
    policies:
    - name: basic-auth-policy
    action:
      pass: schema-registry-backend

---
# Basic Authentication Policy
apiVersion: k8s.nginx.org/v1
kind: Policy
metadata:
  name: basic-auth-policy
  namespace: kite-schema-registry-ns
spec:
  basicAuth:
    secret: schema-auth-secret
    realm: "Schema Registry Admin"


# Notes:
# - Backend protocol is HTTPS (upstream.tls.
enable: true)
# - Basic authentication is enforced via Policy
resource
# - SSL passthrough is NOT used (TLS terminates at
ingress)
# - Auth secret must be in same namespace as
VirtualServer
# - Policy must be in same namespace as
VirtualServer
```

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

Community Nginx Annotation OAuth2

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  annotations:
    kubernetes.io/ingress.class: nginx
    nginx.ingress.kubernetes.io/auth-signin:
https://$host/oauth2/start?rd=$escaped_request_uri
    nginx.ingress.kubernetes.io/auth-url:
http://skywalking-ui.skywalking-ns.svc.cluster.
local:4180/oauth2/auth
  name: skywalking-ingress
  namespace: skywalking-ns
spec:
  rules:
  - host: skywalking.gcp-us-east4-tlmy-prod.carbon.
lowes.com
    http:
      paths:
      - backend:
          service:
            name: skywalking-ui
            port:
              number: 80
        path: /
        pathType: ImplementationSpecific
  tls:
  - hosts:
    - skywalking.gcp-us-east4-tlmy-prod.carbon.
lowes.com
    secretName: lowes-ca-signed-cert
```

F5 Nginx Annotation OAuth2

```
apiVersion: k8s.nginx.org/v1
kind: VirtualServer
metadata:
  name: skywalking-ingress
  namespace: skywalking-ns
  labels:
    argocd.argoproj.io/tracking-id: skywalking:
networking.k8s.io/Ingress:skywalking-ns/skywalking-
ingress
spec:
  host: skywalking.gcp-us-east4-tlmy-prod.carbon.
lowes.com
  tls:
    secret: lowes-ca-signed-cert
  upstreams:
  - name: skywalking-backend
    service: skywalking-ui
    port: 80
  routes:
  - path: /
    action:
      pass: skywalking-backend
    location-snippets: |
      auth_request /oauth2/auth;
      error_page 401 = /oauth2/start?
rd=$escaped_request_uri;
      auth_request_set $user
$upstream_http_x_auth_request_user;
      auth_request_set $email
$upstream_http_x_auth_request_email;
      proxy_set_header X-User $user;
      proxy_set_header X-Email $email;
      auth_request_set $auth_cookie
$upstream_http_set_cookie;
      add_header Set-Cookie $auth_cookie;
  - path: /oauth2
    location-snippets: |
      proxy_pass http://skywalking-ui.skywalking-
ns.svc.cluster.local:4180;
      proxy_set_header Host $host;
      proxy_set_header X-Real-IP $remote_addr;
      proxy_set_header X-Scheme $scheme;
      proxy_set_header X-Auth-Request-Redirect
$request_uri;

---
# Notes:
# This configuration uses external OAuth2
authentication via oauth2-proxy
```

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

Community Nginx Annotations Schema Delete

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  annotations:
    kubernetes.io/ingress.class: nginx
    nginx.ingress.kubernetes.io/backend-protocol:
HTTPS
    nginx.ingress.kubernetes.io/configuration-
snippet: |
      if ($request_method = DELETE) {
        return 403;
      }
  name: test-schema-registry-ingress
  namespace: kite-schema-registry-ns
spec:
  ingressClassName: nginx
  rules:
  - host: test-schema.east4.kite.carbon-dev01.gcp.
lowes.com
    http:
      paths:
      - backend:
          service:
            name: registry-svc
            port:
              number: 8081
        path: /
        pathType: ImplementationSpecific
  tls:
  - hosts:
    - test-schema.east4.kite.carbon-dev01.gcp.
lowes.com
    secretName: test-schema-registry-ext-listener-
cert
```

F5 Nginx Schema Delete

```
apiVersion: k8s.nginx.org/v1
kind: VirtualServer
metadata:
  name: test-schema-registry-ingress
  namespace: kite-schema-registry-ns
spec:
  host: test-schema.east4.kite.carbon-dev01.gcp.
lowes.com
  tls:
    secret: test-schema-registry-ext-listener-cert
  upstreams:
  - name: registry-backend
    service: registry-svc
    port: 8081
    tls:
      enable: true
  routes:
  - path: /
    action:
      pass: registry-backend
    location-snippets: |
      # Block DELETE method - return 403 Forbidden
      if ($request_method = DELETE) {
        return 403;
      }
```

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

Community Nginx Annotations for gRPC

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  annotations:
    kubernetes.io/ingress.class: nginx
    nginx.ingress.kubernetes.io/backend-protocol:
GRPC
    nginx.ingress.kubernetes.io/enable-underscores-
in-headers: "true"
    nginx.ingress.kubernetes.io/use-forwarded-
headers: "true"
  name: tlmy-otel-collector-otel-collector-ingress-
grpc
  namespace: otel-collector-ns
spec:
  ingressClassName: nginx
  rules:
  - host: otel-collector-grpc.gcp-us-east4-tlmy-
prod.carbon.lowes.com
    http:
      paths:
      - backend:
          service:
            name: tlmy-otel-collector-
opentelemetry-collector
            port:
              number: 4317
        path: /
        pathType: Prefix
  tls:
  - hosts:
    - otel-collector-grpc.gcp-us-east4-tlmy-prod.
carbon.lowes.com
    secretName: lowes-ca-signed-cert
```

F5 Nginx for gRPC

```
apiVersion: k8s.nginx.org/v1
kind: VirtualServer
metadata:
  name: tlmy-otel-collector-otel-collector-ingress-
grpc
  namespace: otel-collector-ns
spec:
  host: otel-collector-grpc.gcp-us-east4-tlmy-prod.
carbon.lowes.com
  tls:
    secret: lowes-ca-signed-cert
  upstreams:
  - name: otel-collector-backend
    service: tlmy-otel-collector-opentelemetry-
collector
    port: 4317
    type: grpc
  routes:
  - path: /
    action:
      pass: otel-collector-backend
    location-snippets: |
      underscores_in_headers on;
      proxy_set_header X-Forwarded-For
$proxy_add_x_forwarded_for;
      proxy_set_header X-Forwarded-Proto $scheme;
      proxy_set_header X-Forwarded-Host $host;
      proxy_set_header X-Forwarded-Port
$server_port;

#Notes:
# Custom header handling with location snippet,
Because F5 NGINX VirtualServer doesn't have native
fields for enable-underscores-in-headers and use-
forwarded-headers - you must use raw NGINX
directives via location-snippets or set them
globally in ConfigMap.
# Replaces: nginx.ingress.kubernetes.io/enable-
underscores-in-headers: "true"
# Replaces: nginx.ingress.kubernetes.io/use-
forwarded-headers: "true"
```

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

Community Nginx SSL

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  annotations:
    nginx.ingress.kubernetes.io/force-ssl-
redirect: "true"
    nginx.ingress.kubernetes.io/ssl-passthrough:
"true"
  name: argocd-d3-server
  namespace: argocd
spec:
  ingressClassName: nginx
  rules:
  - host: app.east4.argocd-d3.carbon-sysctl.gcp.
lowes.com
    http:
      paths:
      - backend:
          service:
            name: argocd-d3-server
            port:
              number: 443
        path: /
        pathType: Prefix
```

F5 Nginx SSL

```
apiVersion: k8s.nginx.org/v1
kind: GlobalConfiguration
metadata:
  name: nginx-configuration
  namespace: nginx-ingress
spec:
  listeners:
  - name: tls-passthrough
    port: 443
    protocol: TLS_PASSTHROUGH
---
apiVersion: k8s.nginx.org/v1
kind: TransportServer
metadata:
  name: argocd-d3-server
  namespace: argocd
spec:
  listener:
    name: tls-passthrough
    protocol: TLS_PASSTHROUGH
  host: app.east4.argocd-d3.carbon-sysctl.gcp.
lowes.com
  upstreams:
  - name: argocd-backend
    service: carbon-argocd-server
    port: 443
  action:
    pass: argocd-backend
```

Community Nginx Annotation OAuth Large Buffers

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  annotations:
    kubernetes.io/ingress.class: nginx
    nginx.ingress.kubernetes.io/large-client-
header-buffers: 4 16k
    nginx.ingress.kubernetes.io/proxy-buffer-size:
8k
    nginx.ingress.kubernetes.io/proxy-buffers-
number: "8"
  name: skywalking-oauth-ingress
  namespace: skywalking-ns
spec:
  rules:
  - host: skywalking.gcp-us-east4-tlmy-prod.carbon.
lowes.com
    http:
      paths:
      - backend:
          service:
            name: skywalking-ui
            port:
              number: 4180
        path: /oauth2
        pathType: ImplementationSpecific
  tls:
  - hosts:
    - skywalking.gcp-us-east4-tlmy-prod.carbon.
lowes.com
    secretName: lowes-ca-signed-cert
```

F5 Nginx OAuth Large Buffers

```
apiVersion: k8s.nginx.org/v1
kind: VirtualServer
metadata:
  name: skywalking-oauth-ingress
  namespace: skywalking-ns
spec:
  host: skywalking.gcp-us-east4-tlmy-prod.carbon.
lowes.com
  tls:
    secret: lowes-ca-signed-cert
  upstreams:
  - name: oauth-proxy-backend
    service: skywalking-ui
    port: 4180
    buffering: true
    buffers:
      number: 8
      size: 8k
  routes:
  - path: /oauth2
    action:
      pass: oauth-proxy-backend
    location-snippets: |
      large_client_header_buffers 4 16k;
      proxy_buffer_size 8k;
      proxy_buffers 8 8k;
```

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated

Documentation:

- [F5 NGINX Ingress Controller Documentation](https://docs.nginx.com/nginx-ingress-controller/)
- [Migration Guide](https://docs.nginx.com/nginx-ingress-controller/install/migrate-ingress-nginx/)
- [Annotations Reference](https://docs.nginx.com/nginx-ingress-controller/configuration/ingress-resources/advanced-configuration-with-annotations/)
- [ConfigMap Reference](https://docs.nginx.com/nginx-ingress-controller/configuration/global-configuration/configmap-resource/)

All documents and pages on Confluence are considered Lowe's Confidential unless otherwise stated